

Project Deliverable 1: Data Collection, Cleaning, and Exploration

Dataset Selection and Justification

For this phase of the knowledge discovery process, a comprehensive hospital patient records dataset was selected. The healthcare_dataset.csv file contains 55,500 patient encounters distributed across 15 distinct attributes. These attributes span demographic profiles (e.g., Age, Gender, Blood Type), clinical identifiers (e.g., Medical Condition, Test Results, Medication), and administrative/financial metrics (e.g., Admission Type, Billing Amount, Insurance Provider).

This dataset rigorously satisfies the project constraints by greatly exceeding the minimum threshold of 500 records and 8-10 attributes. Furthermore, it is highly appropriate for data mining because it mirrors the complexities of real-world health informatics. The multifaceted nature of the variables allows for diverse downstream modeling in future deliverables: regression can be applied to predict hospital billing amounts, classification algorithms can be utilized to forecast test results or admission types, and clustering can be deployed to uncover hidden patient segmentation based on length of stay and medical conditions.

Data Cleaning Methodologies

Raw health information systems frequently export data containing administrative errors, redundancies, and formatting inconsistencies that can severely degrade the predictive power of machine learning models. The cleaning protocol implemented for this dataset targets three primary areas of structural degradation.

First, record deduplication was executed. A preliminary scan of the dataset revealed over 500 duplicated patient entries. Redundant observations artificially inflate algorithmic bias during model training by causing the algorithm to over-weight repeated instances; thus, these exact row matches were purged. Next, text standardization was applied. Administrative inputs for patient names contained severe casing inconsistencies (e.g., "Bobby JacksOn"), which were standardized using title-casing protocols to ensure uniformity, a critical step if natural language processing or specific string matching is required later.

Finally, noisy and logically impossible data were addressed. An analysis of the continuous variables revealed that the Billing Amount column contained anomalous negative values (e.g., -2008.49). Because negative hospital bills are domain-inconsistent (representing data entry noise rather than genuine financial variance), these values were isolated and mathematically imputed using the median billing amount of the valid cohort. Additionally, temporal strings (admission

and discharge dates) were converted into operational datetime objects, allowing for the extraction of a new, highly predictive engineered feature: Length of Stay.

Exploratory Data Analysis Framework

Visual exploration serves as the diagnostic phase, revealing the statistical architecture of the data prior to algorithmic modeling. The exploratory data analysis focused on mapping patient demographics and isolating financial and operational trends.

Histograms mapping the frequency distributions of patient age demonstrated a relatively uniform distribution across adult demographics, indicating a balanced dataset not overly skewed toward geriatric or pediatric cohorts. Box plots were deployed to examine the dispersion of Billing Amount across different Medical Conditions. This visualization is critical for identifying whether certain chronic illnesses carry inherently higher financial variances or extreme outliers. Furthermore, bar charts examining the newly engineered "Length of Stay" metric against "Admission Type" (Elective, Urgent, Emergency) provide foundational operational insights. These visualizations directly dictate future modeling parameters; for instance, the lack of extreme age outliers means heavy standard scaling on demographics may be less critical, whereas the broad variance in billing suggests robust regression techniques will be necessary.